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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read the uploaded image
img = cv2.imread('/content/image.jpg')

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Sobel operator along Y-axis
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Convert result to 8-bit
sobel_y = cv2.convertScaleAbs(sobel_y)

# Display Original Image
print("Original Image")
cv2_imshow(img)

# Display Grayscale Image
print("Grayscale Image")
cv2_imshow(gray)

# Display Sobel Y Edge Detection
print("Sobel Y Edge Detection")
cv2_imshow(sobel_y)
```





Grayscale Image

